CS-GY 6763: Lecture 7
Gradient Descent and Projected Gradient Descent

NYU, Prof. Ainesh Bakshi

optimization

Next Unit: Continuous Optimization

Have some function $f : \mathbb{R}^d \to \mathbb{R}$. Want to find $\mathbf{x}^* \in R^d$ such that:

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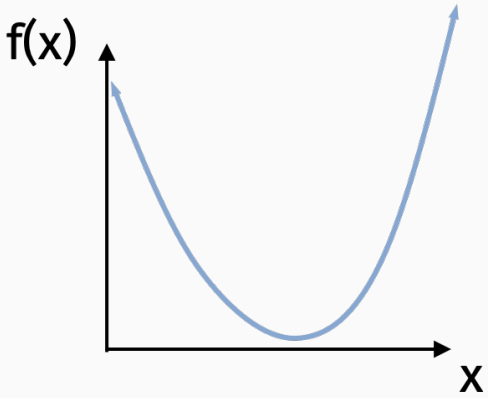
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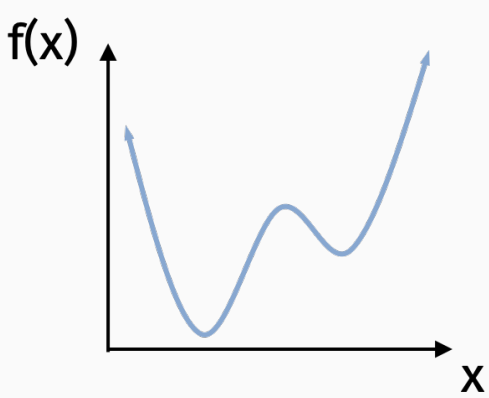
Often we have some additional constraints:

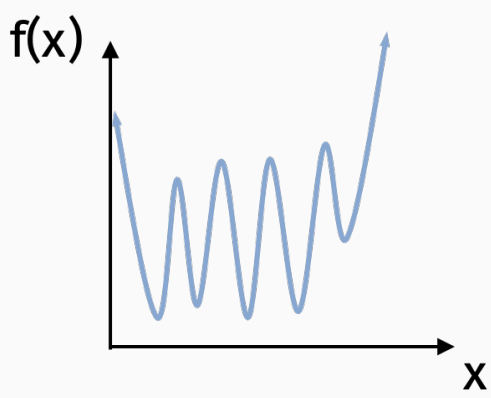
- $\mathbf{x} > 0$. (entry-wise positive)
- $\|\mathbf{x}\|_2 \leq R$, $\|\mathbf{x}\|_1 \leq R$. (convex constraints)
- $\mathbf{a}^T\mathbf{x} = c$. (linear subspace)

Continuous Optimization

Dimension $d = 1$:

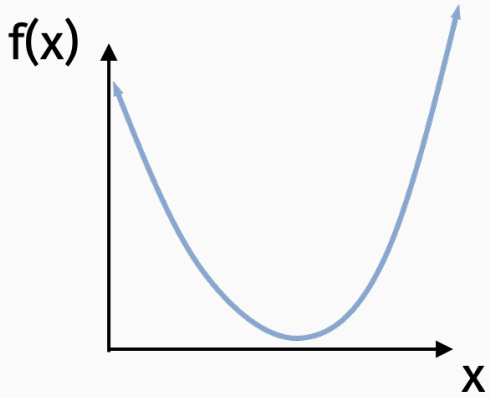


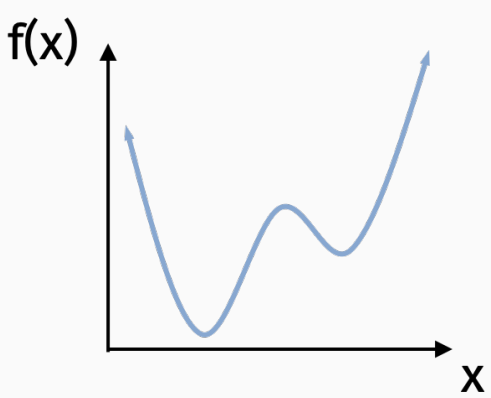


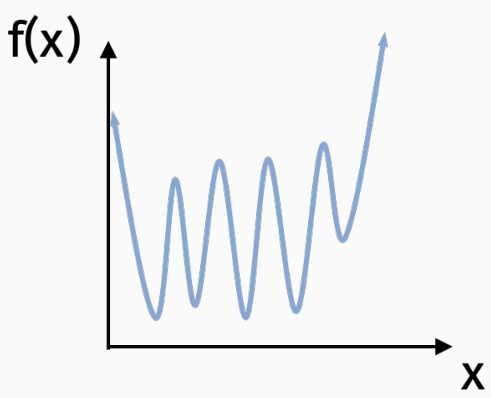


Continuous Optimization

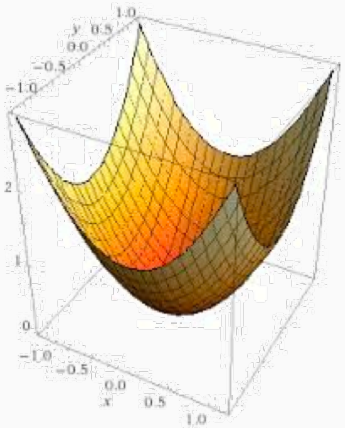
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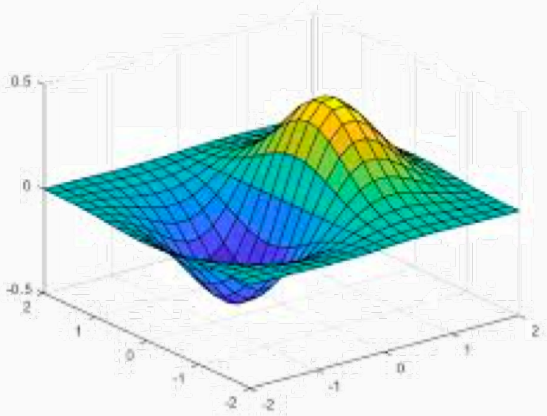


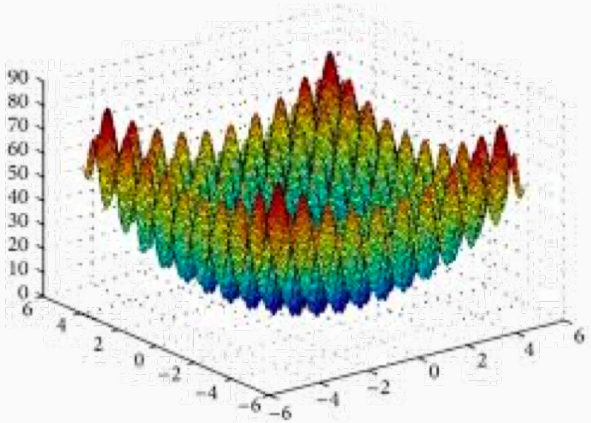




Dimension $d = 2$:







Optimization in Machine Learning

Continuous optimization is the foundation of modern machine learning.

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Supervised learning: Want to learn a model that maps inputs

- numerical data vectors
- images, video
- a sequence of tokens/words

to predictions

- numerical value (probability stock price increases)
- label (does the image contain a car? what is the next token in the sequence?)
- decision (turn car left, rotate robotic arm)

How do we map Supervised Learning back to Continuous Optimization?

Machine Learning Model

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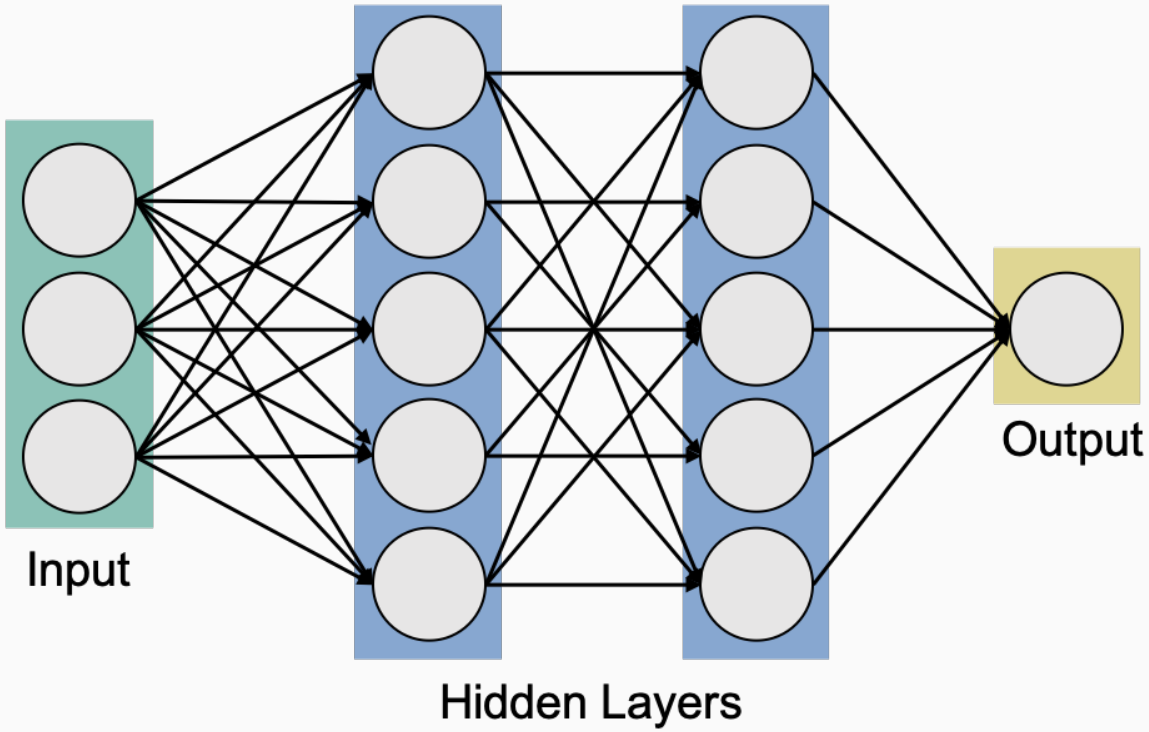
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What model is this?

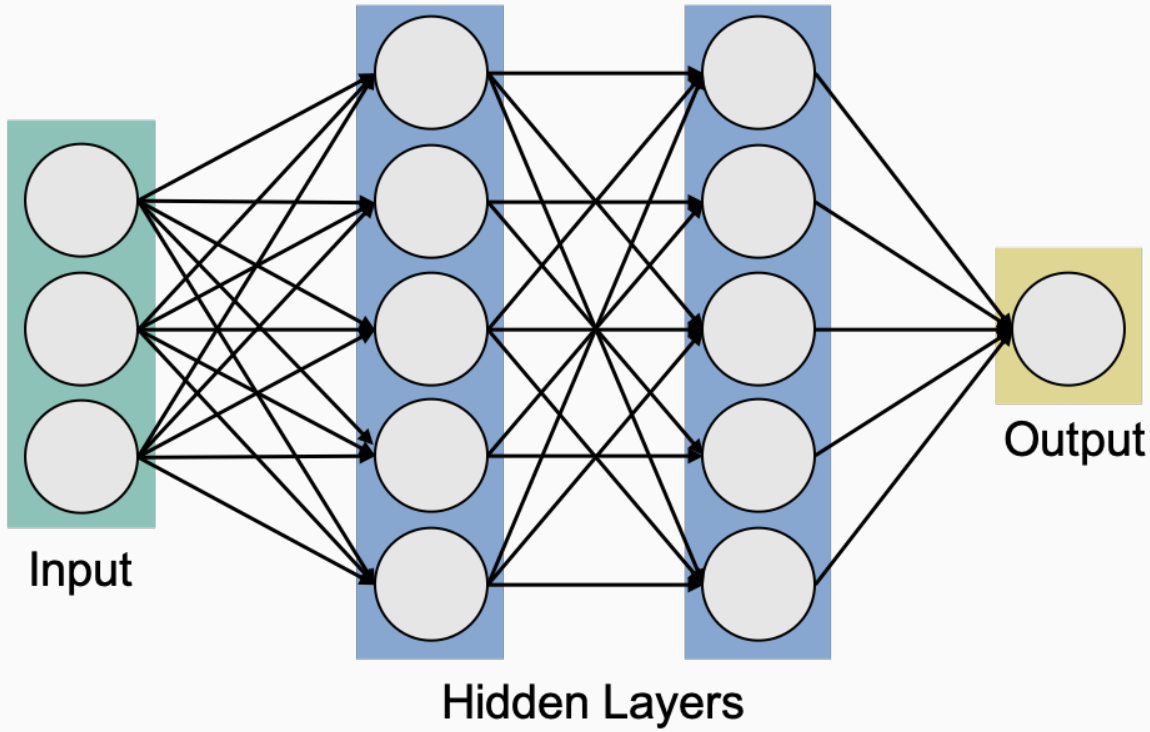
Machine Learning Model

Example: What is the parameter vector **x** here:



Machine Learning Model

Example: What is the parameter vector $\mathbf{x}$ here:



$\mathbf{x} \in \mathbb{R}^{(\#\text{ of connections})}$ is the parameter vector containing all the network weights and biases.

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Want to find $\hat{\mathbf{x}}$ so that $M_{\hat{\mathbf{x}}}(\mathbf{a}^{(i)}) \approx y^{(i)}$ for $i \in 1, \ldots, n$.

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How do we turn this into a function minimization problem?

Loss Function

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- Hinge loss: 1 - $y \cdot M_{\mathbf{x}}(\mathbf{a})$
- Cross-entropy loss (log loss): $-o \log(M_{\mathbf{x}}(\mathbf{y})) + (1 - o) \log(1 - M_{\mathbf{x}}(\mathbf{y}))$

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Solve the optimization problem $\min_{\mathbf{x}} f(\mathbf{x})$.

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Equivalent Reformulation:

$$f(\mathbf{x}) = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$$

where $\mathbf{A}$ is a matrix with $\mathbf{a}^{(i)}$ as its ith row and $\mathbf{y}$ is a vector with $y^{(i)}$ as its ith entry.

Algorithms for Continuous Optimization

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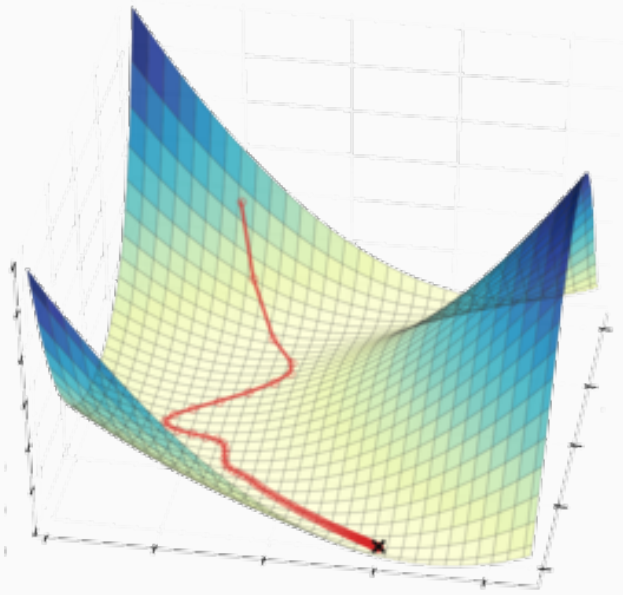
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What are some example algorithms for continuous optimization?

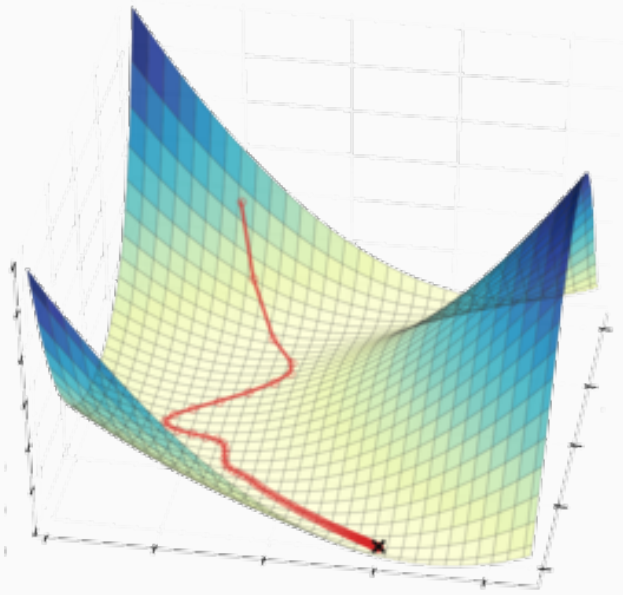
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Runtime generally scales linearly with the dimension of $\mathbf{x}$ (although this is a bit of an over-simplification).

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Faster and more accurate in low-dimensions, slower in very high dimensions. Generally runtime scales polynomially with the dimension of $\mathbf{x}$ (e.g., $O(d^3)$).

Calculus Review

For $i = 1, \ldots, d$, let x_i be the ith entry of $\mathbf{x}$. Let $\mathbf{e}^{(i)}$ be the ith standard basis vector.

Partial derivative:

$$\frac{\partial f}{\partial x_i}(\mathbf{x}) = \lim_{t \to 0} \frac{f(\mathbf{x} + t\mathbf{e}^{(i)}) - f(\mathbf{x})}{t}$$

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The partial derivative for coordinate 1 is the directional derivative along what direction?

Calculus Review

Gradient:

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(\mathbf{x}) \\ \frac{\partial f}{\partial x_2}(\mathbf{x}) \\ \vdots \\ \frac{\partial f}{\partial x_d}(\mathbf{x}) \end{bmatrix}$$

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First Order Optimization

Helpful to think about the first order oracle model for optimization

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- **Gradient oracle**: Evaluate $\nabla f(\mathbf{x})$ for any $\mathbf{x}$.

We view the implementation of these oracles as black-boxes, but they can often require a fair bit of computation.

Example Gradient Evaluation

Linear least-squares regression:

- Given $\mathbf{a}^{(1)}, \ldots \mathbf{a}^{(n)} \in \mathbb{R}^d$, $y^{(1)}, \ldots y^{(n)} \in \mathbb{R}$.
- Want to minimize:

$$f(\mathbf{x}) = \sum_{i=1}^{n} \left(\mathbf{x}^T \mathbf{a}^{(i)} - y^{(i)} \right)^2 = \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2.$$

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$\mathcal{O}(nd)$ (linear in d)

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Leading question: When η is small, what's an approximation for $f(\mathbf{x} + \eta\mathbf{v}) - f(\mathbf{x})$?

$$f(\mathbf{x} + \eta\mathbf{v}) - f(\mathbf{x}) \approx \eta \cdot \nabla f(\mathbf{x})^T \mathbf{v}$$

Directional Derivatives

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How should we choose v so that $f(\mathbf{x} + \eta \mathbf{v}) < f(\mathbf{x})$?

One choice: pick $v = -\nabla f(\mathbf{x})$. Then,

$$\eta \cdot \nabla f(\mathbf{x})^T \mathbf{v} = -\eta \cdot \nabla f(\mathbf{x})^T f(\mathbf{x}) = -\eta \cdot \|\nabla f(\mathbf{x})\|_2^2$$

Gradient Descent

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Gradient Descent

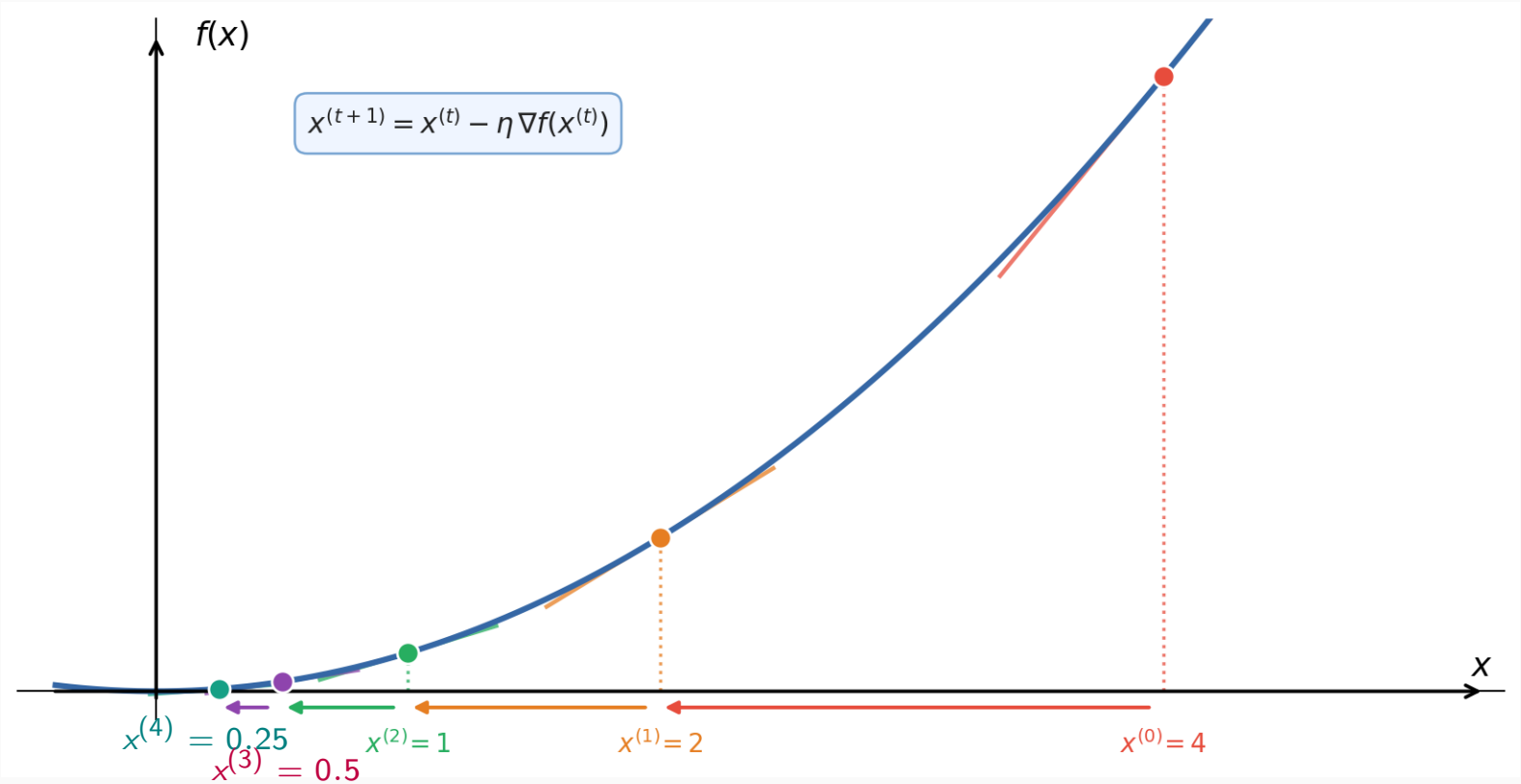
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η is a step-size parameter, which is often adapted on the go. For now, assume it is fixed ahead of time.

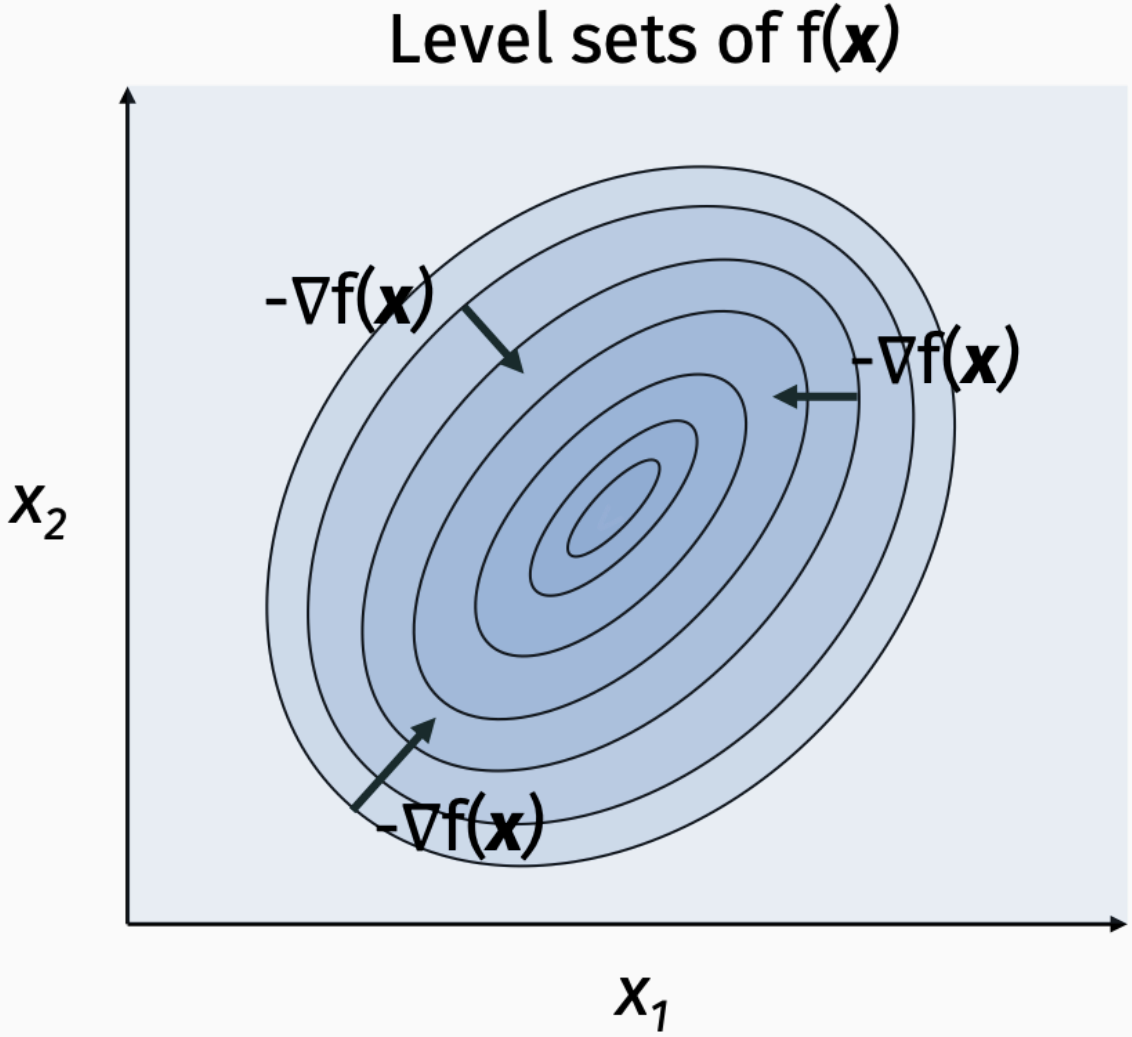
Gradient Descent Intuition

1 dimensional example: $f(x) = x^2$, starting at $x^{(0)} = 4$, step size $\eta = 0.25$.



Gradient Descent Intuition

2 dimensional example:



Key Results

For a convex function $f(\mathbf{x})$: For sufficiently small η and a sufficiently large number of iterations T, gradient descent will converge to a **near global minimum**:

$$f(\mathbf{x}^{(T)}) \leq f(\mathbf{x}^*) + \epsilon.$$

Examples: least squares regression, logistic regression, kernel regression, SVMs.

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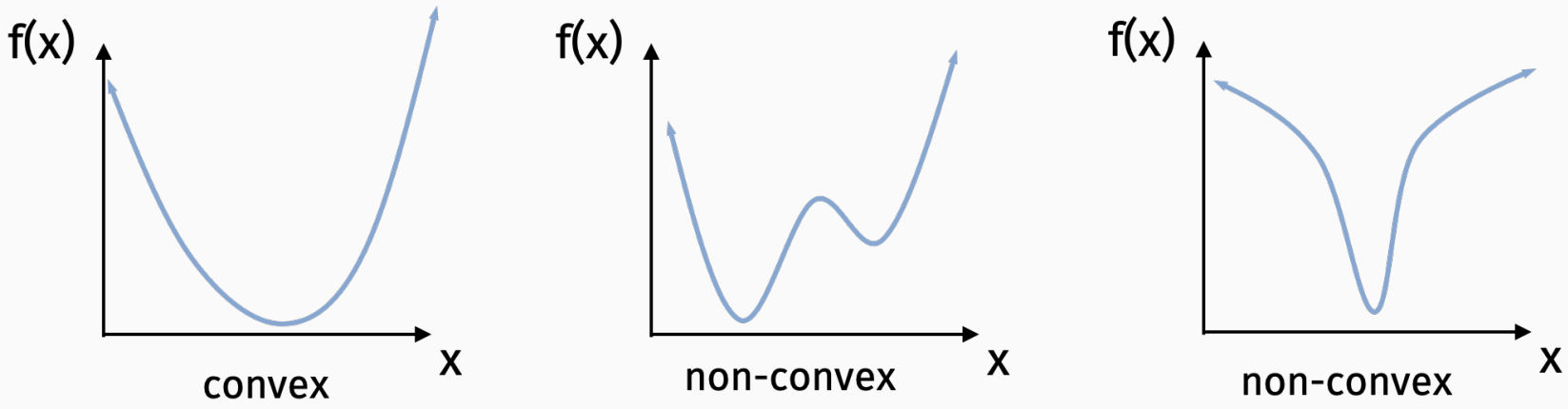
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For a non-convex function $f(\mathbf{x})$: For sufficiently small η and a sufficiently large number of iterations T, gradient descent will converge to a **near stationary point** (stationary point $\nabla f(\mathbf{x}^{(T)}) = 0$):

$$\|\nabla f(\mathbf{x}^{(T)})\|_2 \leq \epsilon.$$

Examples: neural networks, matrix completion problems, mixture models.

Convex vs. Non-Convex



One issue with non-convex functions is that they can have **local minima**. Even when they don't, convergence analysis requires different assumptions than convex functions.

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- Bounding iteration complexity requires placing some assumptions on $f(\mathbf{x})$.
- Stronger assumptions lead to better bounds on the convergence.

Understanding these assumptions can help us design faster variants of gradient descent (there are many!).

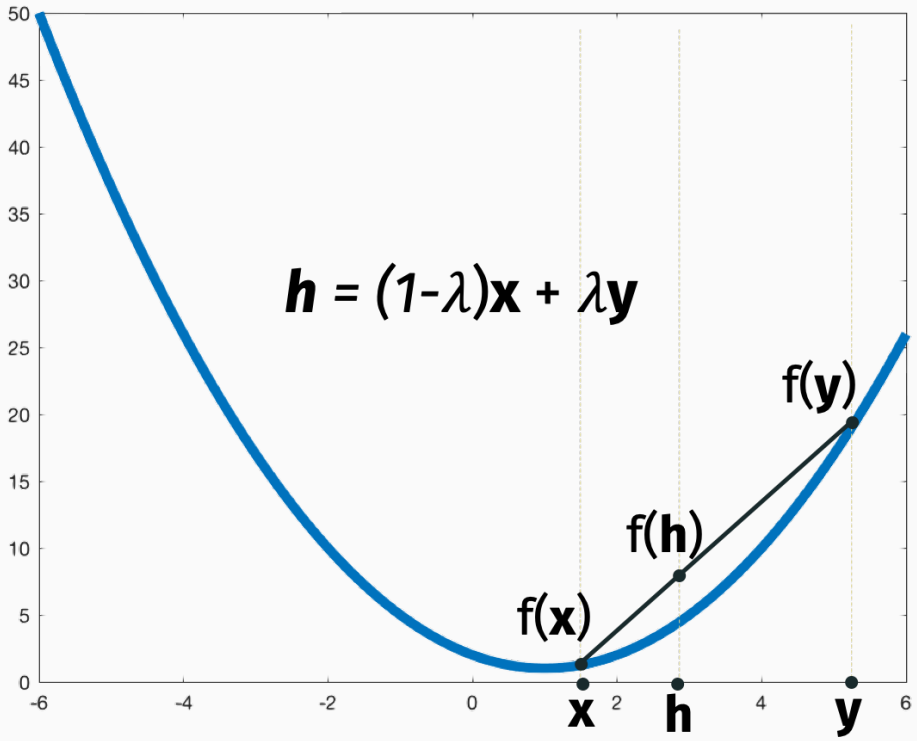
Today, we will start with **convex** functions.

Convexity

Definition (Convex)

A function f is convex iff for any $\mathbf{x}, \mathbf{y}, \lambda \in [0, 1]$:

$$(1 - \lambda) \cdot f(\mathbf{x}) + \lambda \cdot f(\mathbf{y}) \geq f((1 - \lambda) \cdot \mathbf{x} + \lambda \cdot \mathbf{y})$$



Gradient Descent

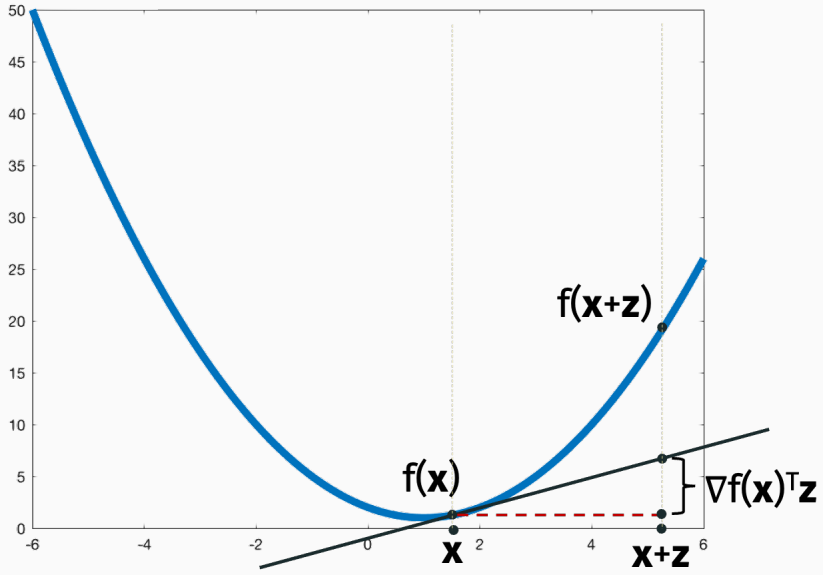
Definition (Convex)

A function f is convex if and only if for any $\mathbf{x}, \mathbf{z}$:

$$f(\mathbf{x}+\mathbf{z}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T \mathbf{z}$$

Equivalently: set $\mathbf{z} = y - x$, and therefore

$$f(\mathbf{x}) - f(\mathbf{y}) \leq \nabla f(\mathbf{x})^T (\mathbf{x} - \mathbf{y})$$



Definitions of Convexity

It is easy but not obvious how to prove the equivalence between chord and tangent definitions. A short proof can be found in Karthik Sridharan's lecture notes here:

http://www.cs.cornell.edu/courses/cs6783/2018fa/lec16-supplement.pdf

Gradient Descent Analysis

Assume:

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- Return $\hat{\mathbf{x}} = \arg\min_{\mathbf{x}^{(i)}} f(\mathbf{x}^{(i)})$. (return the minimum you found along the way)

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Claim (GD Convergence Bound)

If we run GD for $T \geq \frac{R^2 G^2}{\epsilon^2}$ *iterations with step-size* $\eta = \frac{R}{G\sqrt{T}}$, *then* $f(\hat{\mathbf{x}}) \leq f(\mathbf{x}^*) + \epsilon$.

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Of course the best solution found, $\hat{\mathbf{x}}$ is only better than the average.

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Gradient Descent Analysis

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If $T \geq \frac{R^2 G^2}{\epsilon^2}$ *and* $\eta = \frac{R}{G\sqrt{T}}$, *then* $f(\widehat{\mathbf{x}}) \leq f(\mathbf{x}^*) + \epsilon$.

Claim 1: For all $i = 0, \ldots, T$,

$$f(\mathbf{x}^{(i)}) - f(\mathbf{x}^*) \leq \frac{\|\mathbf{x}^{(i)} - \mathbf{x}^*\|_2^2 - \|\mathbf{x}^{(i+1)} - \mathbf{x}^*\|_2^2}{2\eta} + \frac{\eta G^2}{2}$$

Telescoping sum:

$$\begin{aligned}
\sum_{i=0}^{T-1} \left[f(\mathbf{x}^{(i)}) - f(\mathbf{x}^*) \right] \leq & \frac{\|\mathbf{x}^{(0)} - \mathbf{x}^*\|_2^2 - \|\mathbf{x}^{(1)} - \mathbf{x}^*\|_2^2}{2\eta} + \frac{\eta G^2}{2} \\
& + \frac{\|\mathbf{x}^{(1)} - \mathbf{x}^*\|_2^2 - \|\mathbf{x}^{(2)} - \mathbf{x}^*\|_2^2}{2\eta} + \frac{\eta G^2}{2} \\
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& \vdots \\
& + \frac{\|\mathbf{x}^{(T-1)} - \mathbf{x}^*\|_2^2 - \|\mathbf{x}^{(T)} - \mathbf{x}^*\|_2^2}{2\eta} + \frac{\eta G^2}{2}
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Dividing both sides by T,

$$\frac{1}{T} \sum_{i=0}^{T-1} \left[f(\mathbf{x}^{(i)}) - f(\mathbf{x}^*) \right] \leq \frac{RG}{\sqrt{T}} \leq \epsilon$$

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$$\implies \left[\frac{1}{T}\sum_{i=0}^{T-1} f(\mathbf{x}^{(i)})\right] - f(\mathbf{x}^*) \leq \epsilon$$

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We always have that $f(\hat{\mathbf{x}}) = \min_i f(\mathbf{x}^{(i)}) \leq \frac{1}{T}\sum_{i=0}^{T-1} f(\mathbf{x}^{(i)})$, which gives the final bound:

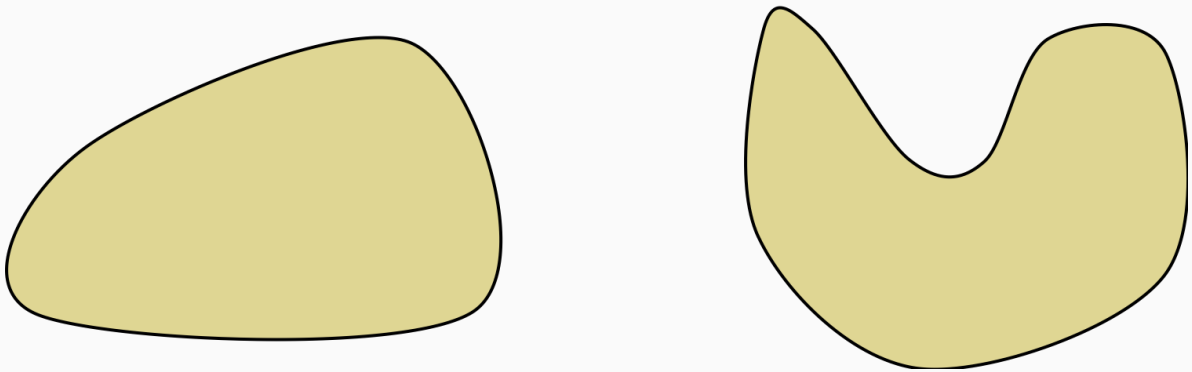
$$f(\hat{\mathbf{x}}) \leq f(\mathbf{x}^*) + \epsilon.$$

Constrained Convex Optimization

Typical goal: Solve a convex minimization problem with additional convex constraints.

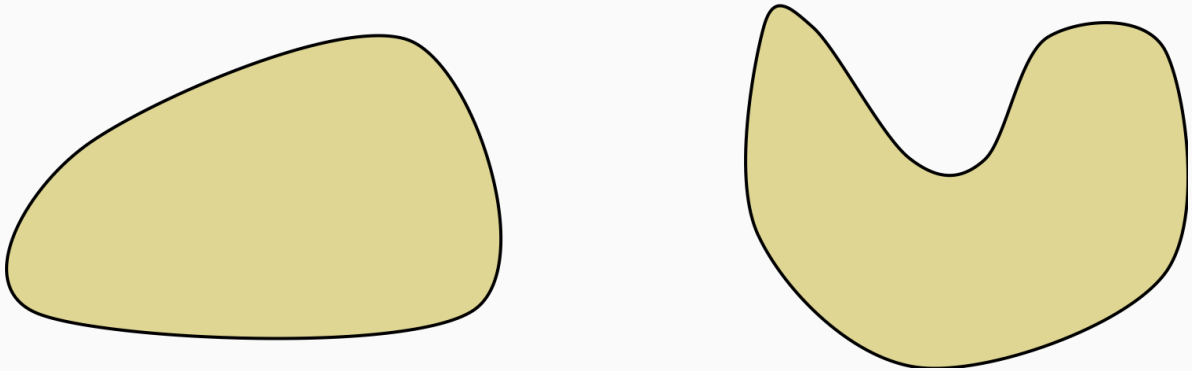
$$\min_{\mathbf{x} \in \mathcal{S}} f(\mathbf{x})$$

where $\mathcal{S}$ is a **convex set**.



Which of these is convex?

Constrained Convex Optimization



Definition (Convex set)

A set $\mathcal{S}$ is convex if for any $\mathbf{x}, \mathbf{y} \in \mathcal{S}, \lambda \in [0, 1]$:

$$(1 - \lambda)\mathbf{x} + \lambda\mathbf{y} \in \mathcal{S}.$$

Constrained Convex Optimization

Examples:

- **Norm constraint:** minimize $\|\mathbf{Ax} - \mathbf{b}\|_2$ subject to $\|\mathbf{x}\|_2 \leq \lambda$. Used e.g. for regularization, finding a sparse solution, etc.

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- **Positivity constraint:** minimize $f(\mathbf{x})$ subject to $\mathbf{x} \geq 0$.

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Examples:

- **Norm constraint:** minimize $\|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2$ subject to $\|\mathbf{x}\|_2 \leq \lambda$. Used e.g. for regularization, finding a sparse solution, etc.

- **Positivity constraint:** minimize $f(\mathbf{x})$ subject to $\mathbf{x} \geq 0$.

- **Linear constraint:** minimize $\mathbf{c}^T\mathbf{x}$ subject to $\mathbf{A}\mathbf{x} \leq \mathbf{b}$. Linear program used in training support vector machines, industrial optimization, subroutine in integer programming, etc.

Problem with Gradient Descent

Gradient descent:

- For $i = 0, \ldots, T$:
  - $\mathbf{x}^{(i+1)} = \mathbf{x}^{(i)} - \eta \nabla f(\mathbf{x}^{(i)})$
- Return $\hat{\mathbf{x}} = \arg\min_i f(\mathbf{x}^{(i)})$.

Even if we start with $\mathbf{x}^{(0)} \in \mathcal{S}$, there is no guarantee that $\mathbf{x}^{(0)} - \eta \nabla f(\mathbf{x}^{(0)})$ will remain in our set.

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Extremely simple modification: Force $\mathbf{x}^{(i)}$ to be in $\mathcal{S}$ by **projecting** onto the set.

Constrained First Order Optimization

Given a function f to minimize and a convex constraint set $\mathcal{S}$, assume we have:

- **Function oracle**: Evaluate $f(\mathbf{x})$ for any $\mathbf{x}$.
- **Gradient oracle**: Evaluate $\nabla f(\mathbf{x})$ for any $\mathbf{x}$.
- **Projection oracle**: Evaluate $P_{\mathcal{S}}(\mathbf{x})$ for any $\mathbf{x}$.

$$P_{\mathcal{S}}(\mathbf{x}) = \arg\min_{\mathbf{y} \in \mathcal{S}} \|\mathbf{x} - \mathbf{y}\|_2$$

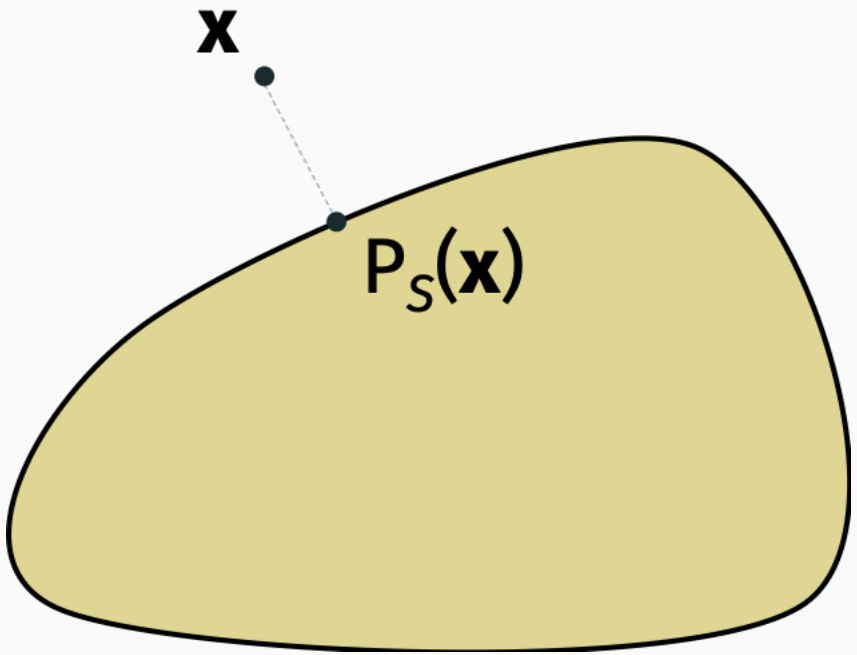
Projection Oracles

- How would you implement $P_{\mathcal{S}}$ for $\mathcal{S} = \{\mathbf{y} : \|\mathbf{y}\|_2 \leq 1\}$.

Projection Oracles

- How would you implement $P_{\mathcal{S}}$ for $\mathcal{S} = \{\mathbf{y} : \|\mathbf{y}\|_2 \leq 1\}$.
- How would you implement $P_{\mathcal{S}}$ for $\mathcal{S} = \{\mathbf{y} : \mathbf{y} = \mathbf{Q}\mathbf{z}\}$, where $Q \in \mathbb{R}^{n \times k}$

$$z^* = \arg\min_{z \in \mathbb{R}^k} \|Qz - x\|_2^2, \qquad \text{thus } z^* = (Q^T Q)^{-1} Q^T x$$



Projected Gradient Descent

Given function $f(\mathbf{x})$ and set $\mathcal{S}$, such that $\|\nabla f(\mathbf{x})\|_2 \leq G$ for all $\mathbf{x} \in \mathcal{S}$ and starting point $\mathbf{x}^{(0)}$ with $\|\mathbf{x}^{(0)} - \mathbf{x}^*\|_2 \leq R$.

Projected gradient descent:

- Select starting point $\mathbf{x}^{(0)}$, $\eta = \frac{R}{G\sqrt{T}}$.
- For $i = 0, \ldots, T$:
  - $\mathbf{z} = \mathbf{x}^{(i)} - \eta \nabla f(\mathbf{x}^{(i)})$
  - $\mathbf{x}^{(i+1)} = P_{\mathcal{S}}(\mathbf{z})$
- Return $\hat{\mathbf{x}} = \arg\min_i f(\mathbf{x}^{(i)})$.

Claim (PGD Convergence Bound)

If $f, \mathcal{S}$ are convex and $T \geq \frac{R^2 G^2}{\epsilon^2}$, then $f(\hat{\mathbf{x}}) \leq f(\mathbf{x}^*) + \epsilon$.

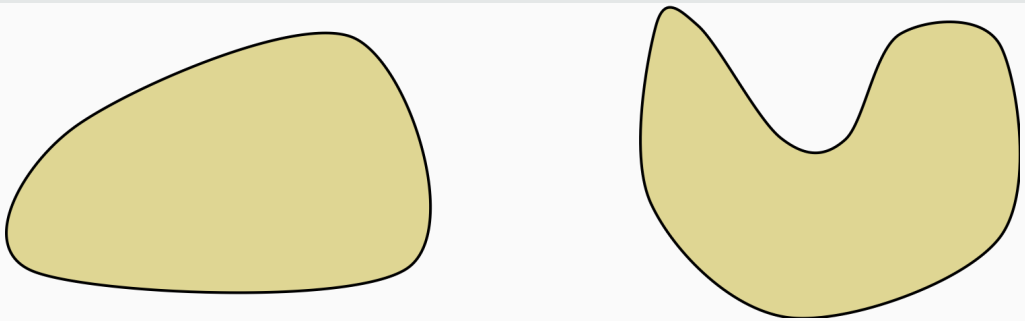
Projected Gradient Descent Analysis

Analysis is almost identical to standard gradient descent! We just need one additional claim:

Claim (Contraction Property of Convex Projection)

If $\mathcal{S}$ is convex, then for any $\mathbf{y} \in \mathcal{S}$,

$$\|\mathbf{y} - P_{\mathcal{S}}(\mathbf{x})\|_2 \leq \|\mathbf{y} - \mathbf{x}\|_2.$$



$$\|\mathbf{y} - \mathbf{x}\|_2^2 = \|\mathbf{y} - \mathbf{x} \pm P_{\mathcal{S}}(\mathbf{x})\|_2^2$$

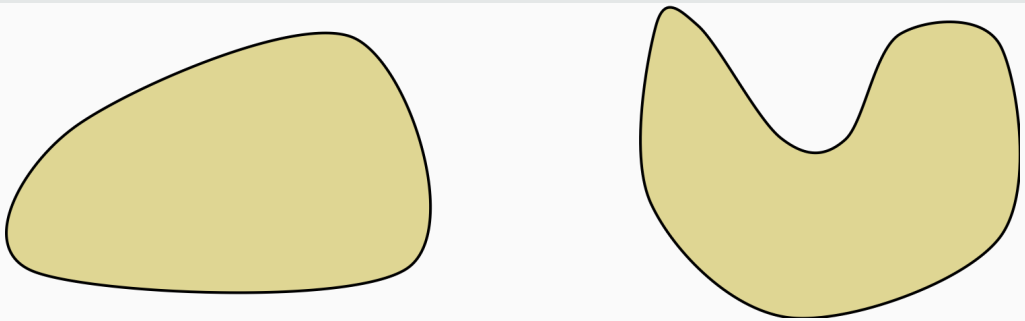
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$$\begin{aligned}\|\mathbf{y} - \mathbf{x}\|_2^2 &= \|\mathbf{y} - \mathbf{x} \pm P_{\mathcal{S}}(\mathbf{x})\|_2^2 \\ &= \|\mathbf{x} - P_{\mathcal{S}}(\mathbf{x})\|_2^2 + \|\mathbf{y} - P_{\mathcal{S}}(\mathbf{x})\|_2^2 + 2\underbrace{\langle \mathbf{x} - P_{\mathcal{S}}(\mathbf{x}), P_{\mathcal{S}}(\mathbf{x}) - \mathbf{y}\rangle}_{\geq 0}\end{aligned}$$

Gradient Descent Analysis

Claim (PGD Convergence Bound)

If $f, \mathcal{S}$ are convex and $T \geq \frac{R^2 G^2}{\epsilon^2}$, then $f(\hat{\mathbf{x}}) \leq f(\mathbf{x}^*) + \epsilon$.

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Same telescoping sum argument:

$$\left[\frac{1}{T}\sum_{i=0}^{T-1} f(\mathbf{x}^{(i)})\right] - f(\mathbf{x}^*) \leq \frac{R^2}{2T\eta} + \frac{\eta G^2}{2}.$$

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Gradient Descent

Conditions:

- **Convexity:** f is a convex function, $\mathcal{S}$ is a convex set.
- **Bounded Initial Distance:**

$$\|\mathbf{x}^{(0)} - \mathbf{x}^*\|_2 \leq R$$

- **Bounded gradients (Lipschitz function)**:

$$\|\nabla f(\mathbf{x})\|_2 \leq G \text{ for all } \mathbf{x} \in \mathcal{S}.$$

Theorem (GD Convergence Bound)

(Projected) Gradient Descent outputs $\hat{\mathbf{x}}$ *with*

$$f(\hat{\mathbf{x}}) \leq \min_{\mathbf{x}\in\mathcal{S}} f(\mathbf{x}) + \epsilon\,, \qquad after \quad T = \frac{R^2 G^2}{\epsilon^2} \text{ iterations.}$$